

Problem 2

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Problem 1

(a) Fill in the blanks

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

if the limit exists.

Explanation.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

if the limit exists.

(b) Let

$$f(x) = \frac{1}{x+4}.$$

Use the (limit) definition of derivative in (a) to find $f'(x)$.

Explanation.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h+4} - \frac{1}{x+4}}{h} \text{ which has form } \frac{0}{0} \\ &= \lim_{h \rightarrow 0} \frac{\frac{x+4 - (x+h+4)}{(x+h+4)(x+4)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{-h}{(x+h+4)(x+4)} \\ &= \lim_{h \rightarrow 0} \frac{-1}{(x+h+4)(x+4)} = \frac{-1}{(x+4)^2} \end{aligned}$$

[3]